

Enabling Large Scale LoRa Parallel Decoding With High-Dimensional and High-Accuracy Features

Weiwei Chen[✉], Xianjin Xia[✉], *Member, IEEE*, Shuai Wang[✉], Tian He[✉], *Fellow, IEEE*, Shuai Wang[✉], Gang Liu[✉], and Caishi Huang[✉]

Abstract—LoRaWAN is a prominent technology for Low Power Wide Area Networks (LPWAN). However, the increasing network size has introduced a significant challenge: packet collisions resulting from concurrent transmissions in LoRaWAN. Previous studies either overlooked the issue by examining limited features or tackled it with intricate receivers employing up to eight antennas. To achieve a more favorable balance between implementation cost and system performance, we introduce Hi²LoRa—a solution utilizing highly dimensional and accurate features for LoRa concurrent decoding, implemented with only two receiving antennas. The feature dimensions are expanded through an exploration of various hardware imperfections and inherent channel state information specific to each transceiver pair. To enhance feature accuracy, low pass filters and BiLSTM networks are applied to capture and learn their temporal patterns. Additionally, an efficient collision suppression strategy is introduced to mitigate feature corruption from concurrently transmitted packets. Extensive real-world testbed evaluations demonstrate that the achievable concurrency in Hi²LoRa approaches that of state-of-the-art approaches with significantly higher complexity (e.g., utilizing eight antennas) or exceeds prior work by a factor of 2.7 with comparable complexity (e.g., using two antennas).

Index Terms—LoRaWAN, parallel decoding, multiple antennas, interference suppression.

I. INTRODUCTION

OVER the past few years, Low Power Wide Area (LPWA) networks have experienced rapid development. According to Cisco's Virtual Network Index (VNI) [1], LPWA connections are expected to grow to 1.9 billion by 2023. Among the various LPWA technologies, LoRaWAN (LoRa Wide Area Network) [2], [3] has gained significant attention worldwide.

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Weiwei Chen is with the School of Computer Engineering and Science, Shanghai University, Shanghai 200444, China (e-mail: chen.ava.0012@gmail.com).

Xianjin Xia is with the Department of Computer Science, The Hong Kong Polytechnic University, Hong Kong.

Shuai Wang, Tian He, and Shuai Wang are with the School of Computer Science and Engineering, Southeast University, Nanjing 211102, China.

Gang Liu is with the Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Sichuan 610056, China.

Caishi Huang is with the Department of Biomedical Science, University of Macau, Macau 999078, China.

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As LoRaWAN utilizes a lightweight ALOHA-based MAC protocol [4], the increasing network size makes concurrent transmissions inevitable. If this issue is not adequately addressed, it can significantly affect network performance.

Prior work has leveraged signal features from both the time and frequency domains at the PHY layer to decode concurrent packets. Specifically, Ftrack [5] and CIC [6] examine the frequency continuity and timing misalignment of chirp signals to separate concurrent symbols. CoLoRa [7] and PCube [8] explore amplitude or phase features to decode symbols in concurrent packets. SCLoRa [9], NScale [10], and Pyramid [11] jointly consider timing and amplitude features of the signals for concurrent decoding. Magnifier [12] investigates multiple hardware imperfections for parallel decoding. However, although these approaches perform well for low-concurrency networks, we argue that they are inefficient in handling large-scale concurrent transmissions.

The LoRa physical layer (PHY) partitions a channel into multiple evenly spaced frequency intervals, referred to as *frequency bins*. Each frequency bin is allocated to a specific symbol. Due to hardware imperfections, the demodulated frequency of a received symbol, termed as the *frequency tone* may deviate from its transmitted frequency by a fractional amount of frequency bin. The receiver quantizes the frequency tone to the frequency bin it locates to decode the symbol. Since the number of available frequency bins is fixed and limited, as the number of concurrent transmitters increases, chances are that more than one concurrent symbols (from different transmitters) using adjacent frequency tones (known as *frequency tone collisions*). This situation is akin to the pigeonhole problem, in which symbols from concurrent packets represent the pigeons and frequency bins encompass the frequency tones are the holes (as seen in Fig. 1(a)). Frequency tone collision brings in substantial feature distortions to the collided frequency tones, making them impossible to be identified (as depicted in Fig. 1(b)). Moreover, the probability that frequency tone collision occurs increases with network concurrency. As shown in Fig. 1(c), with 40 SF10 concurrent packets, the probability of frequency tone collisions reaches 23.7%, leading to unbearable decoding errors. This situation only worsens with more concurrent nodes. Unfortunately, previous studies have not tackled this issue adequately in a simple-to-implement manner.

Besides the frequency tone collisions, previous studies use few features, such as packet amplitude or arrival time, for concurrent decoding. These low-dimensional features restrict

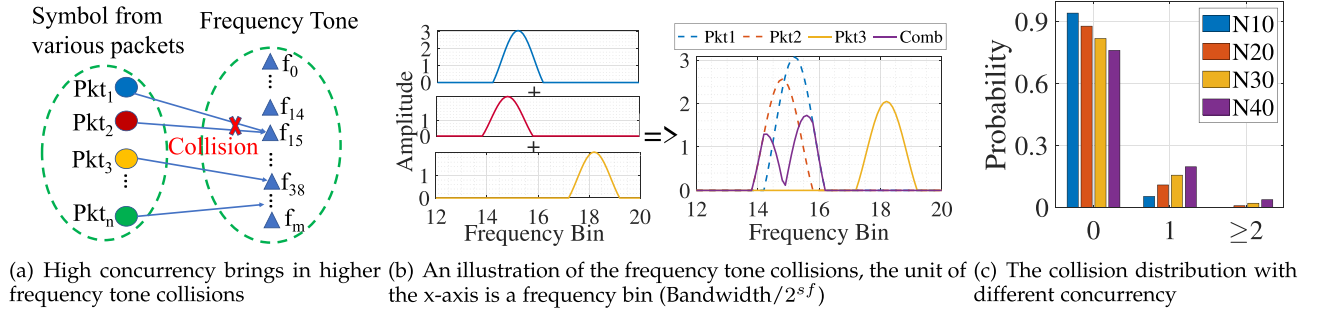


Fig. 1. An illustration of how frequency tone collision happens. In Fig. 1(a), the left column represents the concurrently transmitted symbols from different packets (i.e., from Pkt1, Pkt2, ..., Pktn). Different symbols in a packet will be carried by different frequency tones. In Fig. 1(b), when receiving symbols from three concurrent packets ("Pkt1", "Pkt2" and "Pkt3"), features from the symbols belonging to "Pkt1" and "Pkt2" cannot be retained as their frequency tones collide. The blue line in the figure represents the overall received signals from three concurrent symbols. Due to the collision, features of "Pkt1" and "Pkt2" deviate drastically from their actual values. In Fig. 1(c), "N10" refers to the case with 10 concurrent nodes. All the nodes in Fig. 1(c) transmit SF10 packets.

the network's decoding capacity. When a large number of nodes transmit concurrently, some packets inevitably have close feature values in terms of packet timing, amplitude or both, which can cause ambiguity issues and lead to decoding errors. To tackle this issue, adding more features can help unravel tangled packets. Overall, the features' dimension and accuracy are critical for network concurrency.

To this end, we propose Hi²LoRa, a framework that enables highly dimensional and highly accurate features for LoRa concurrent decoding. Hi²LoRa expands the feature dimension to account for hardware imperfections and the channel state intrinsic to each transceiver pair. Specifically, in Hi²LoRa, the profile of a packet refers to the set of features associated with a packet. This profile is constructed from the offsets, such as carrier frequency offsets (CFO), sampling time offsets (STO), and sampling frequency offsets (SFO), as well as the amplitudes and phases of each frequency tone in a packet. Hi²LoRa decodes the packet by selecting frequency tones whose features match the packet profile. To overcome frequency tone collisions, Hi²LoRa uses two antennas to suppress the strongest interference from one of the colliding symbols.

However, suppressing collisions is a non-trivial task as accurate channel state information is difficult to retrieve when multiple packets are present concurrently. Furthermore, as depicted in Fig. 2 in Section II, the offsets also vary with time, adding to the complexity of the decoding process. To maintain highly precise time-varying packet profiles, we classify the profile into two categories: the channel-insensitive profile, such as offsets, and the channel-sensitive profile, such as amplitude and phase. Hi²LoRa first uses low-pass filters to track the channel-insensitive profile as it changes much slower than the channel-sensitive profile. It then utilizes partial channel-insensitive profiles and deep learning techniques to establish channel-sensitive profiles. Additionally, to suppress frequency tone collisions, channel state information will be constructed from the channel-sensitive profiles. Although Hi²LoRa is tailored for LoRaWAN, the proposed strategies can be generally applied to other IoT systems such as ZigBee and Bluetooth networks to support concurrent transmissions. It can also be utilized to remove Cross Technology Interference (CTI), such

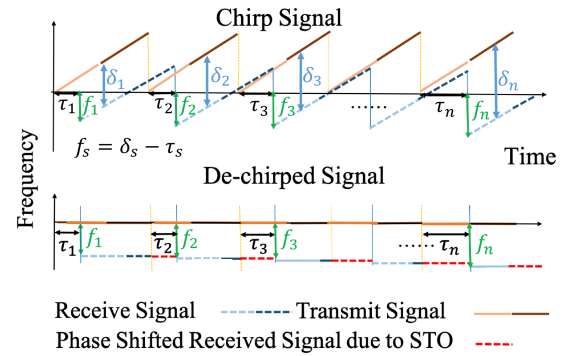


Fig. 2. An illustration of the impacts of offsets on the received signal.

as Wi-Fi's interference with LoRa, ZigBee, or Bluetooth. The contributions are summarized below.

- 1) We propose Hi²LoRa, a cost-effective framework that utilizes highly dimensional and highly accurate features for concurrent LoRa decoding. By effectively suppressing frequency tone collisions, the accuracy of the features can be significantly improved. Moreover, the introduction of high dimensional features can considerably reduce decoding ambiguity, which increases with network concurrency.
- 2) Hi²LoRa utilizes a two-stage algorithm. In the first stage, frequency tones with unambiguous offset features are matched to the packet. The features of these already matched symbols are leveraged to finely align the receiving window to the transmitting signal. The aligned signals are then used to learn and predict the entire packet profile using Bidirectional-Long-Short-Term-Memory (BiLSTM) networks. With the complete profile, symbols yet to be decoded are identified with amplitude, phase and offset information. The strongest interference is suppressed before decoding symbols that face frequency tone collisions.
- 3) We implemented Hi²LoRa on a testbed comprising of 40 off-the-shelf LoRa nodes and a USRP-based gateway with two antennas. Extensive evaluations are conducted, and the results demonstrate that Hi²LoRa achieves a good balance between implementation cost and performance.

Specifically, the achievable concurrency in Hi²LoRa is close to that of PCube [8] with eight antennas, and it achieves up to $2.7\times$ of concurrency compared to PCube with two antennas. Similarly, Hi²LoRa shows up to $3.5\times$ of concurrency compared to CIC [6] with one antenna.

The rest of the paper is organized as follows. Section II provides backgrounds and motivations. Section III presents the overall design structure of Hi²LoRa. Section IV introduces the detailed designs of the Packet Management module in Hi²LoRa. Section V illustrates how the payload of a packet is decode in Hi²LoRa. Section VI demonstrates the test-bed and the evaluation results obtained from it. Related works are studied in Section VII. A conclusion is drawn in Section VIII.

II. BACKGROUNDS AND MOTIVATIONS

A. A Primer on LoRa

LoRa uses CSS (Chirp Spread Spectrum) [13] as its modulation scheme. CSS divides the bandwidth into m equally spaced frequency bins. Each symbol $z_s \in [0, m - 1]$ is modulated by setting the starting frequency to frequency bin z_s . When the signal is received and de-chirped, it is translated into a single frequency tone, which is then quantized to its nearest frequency bin. The frequency bin corresponds to the discrete interval in which a frequency tone is located. Different symbols use different frequency tones, and the presence or absence of a frequency tone encodes the information to be transmitted.

B. Offsets in a Received LoRa Packet

Hardware imperfections can cause offsets between a LoRa transceiver pair. There are three main offsets: CFO (Carrier Frequency Offset), STO (Sampling Time Offset), and SFO (Sampling Frequency Offset). CFO adds frequency drifts in the received signal; STO, besides drifting the frequency, also causes phase shifts of the chips in a symbol. SFO changes the sampling time of every chip in the receiving packet, leading to a time-varying STO for different symbols.

Fig. 2 provides a pictorial representation of the impact of offsets on the received LoRa signal. The upper section of the figure (denoted as Chirp Signal) displays the original chips of every chirp where a set of preambles are transmitted, and each chirp is made up of two chips represented by dark and light lines. The STO, CFO and the overall frequency drift of the s th chirp are denoted by τ_s , δ_s and f_s , respectively, while the difference of the two slopes represents the SFO between the transceivers.

As shown in the figure, the slope of the received signal is lower than that of the transmitted signal, leading to the sampling time at the receiver to lag behind, causing a growing STO over time due to SFO. The lower section of the figure (denoted as De-chirped Signal) shows the de-chirped chips of the preambles. The combination of CFO, STO, and SFO causes the received frequency to deviate from the transmitted one. Moreover, STO also introduces symbol-dependent phase shifts to a subset of chips in a chirp (i.e., the red dash lines in the figure). Overall, the offsets create both frequency and phase drifts when receiving a LoRa packet, leading to a reduction in signal quality.

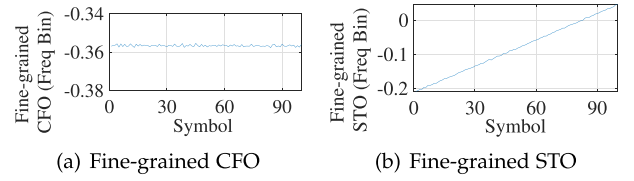


Fig. 3. The fine-grained CFO and the STO of a packet. Here fine-grained CFO means the fractional frequency bin of the CFO. The signals are collected from a SF10 packet transmitted from a COTS SX1278 with bandwidth = 250 KHz.

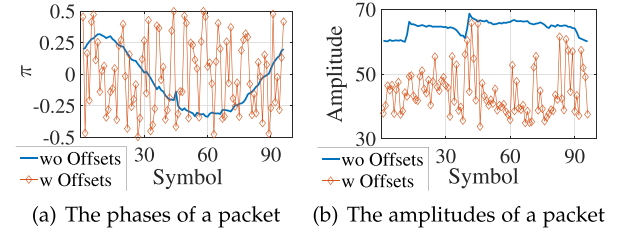


Fig. 4. Phase and amplitude of a packet w/o offsets compensation.

C. Design Opportunities

1) *Hardware Offsets Facilitating the Retrieval of High-Dimensional and High-Accuracy Features:* In LoRa parallel decoding, concurrent packets can be efficiently separated only when a large number of highly accurate features associated with each packet are available. Exploring a variety of offsets in parallel decoding helps to retrieve such features. The reasons are listed as follows.

First, hardware offsets are useful features that can be exploited to expand the feature dimension of each packet. As shown in Choir [14], the cumulative distribution (CDF) of the fractional part of CFO from 30 nodes span across $[-0.5, 0.5]$ frequency bin with equal probability. This observation makes CFO suitable to identify packets from different nodes. In addition, the STO between different packets (even from the same device) are different as the transceiver radio pair is unsynchronized, which triggers a different sampling timing offsets when start transmitting a packet.

Second, hardware offsets can be leveraged to recover other features, such as amplitude and phase features of symbols in a packet with high accuracy. For one thing, features extracted from signals with offset compensation can be accurately estimated and used for parallel decoding. Fig. 3 showcases the CFO and STO of various symbols within a packet. While the CFO is relatively stable, the figure also reveals a gradual increase in STO from -0.2 to 0.05 frequency bins. Complementing this, Fig. 4 demonstrates the amplitude and phase of frequency tones in the same packet both with and without offsets. Notably, the figure reveals large variations in amplitude and phase in a packet when offsets are not compensated. As a result, phase and amplitude features are hard to be estimated accurately without removing the inherent hardware offsets.

To summarize, given various hardware offsets, we can extract eight features from two receiving antennas: CFO (Carrier Frequency Offset), STO (Symbol Timing Offset), phase

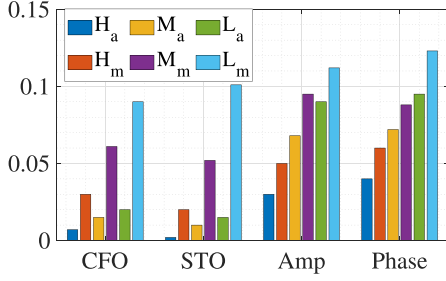


Fig. 5. The average and maximum estimation error for High, Medium and Low SNR symbols.

offset, and bin amplitude for each antenna. In contrast, prior research typically employs a lower-dimensional set of features, such as using only the CFO [14] or time misalignment [5], or combining the amplitude of a frequency bin with frequency continuity [6], and occasionally pairing bin amplitude with time misalignment [9]. Thus, Hi²LoRa surpasses existing methods by leveraging a richer, higher-dimensional feature set.

Additionally, compensating for the offsets help us to extract “high accuracy” features for parallel decoding. We conducted an experiment to evaluate the precision of the fine-grained features for parallel decoding in scenarios involving 40 concurrent nodes. The experiment involved collecting 1000 packets, each containing 50 symbols, from a single node with an average SNR of 50 dB received across two antennas. We computed the associated CFO, STO, amplitude, and phase of each symbol in the packets. Given the minimal noise environment, these computed features were treated as ground-truth values, and the corresponding symbols were labeled as target symbols. To simulate varying noise levels, we recorded environmental noise in the absence of packet transmissions and scaled the energy of the target symbols to match this noise, creating three SNR categories: i) High SNR: SNR > 10 dB; ii) Medium SNR: 0 dB < SNR < 10 dB; and iii). SNR < 0 dB.

For each SNR level, we generated 50,000 symbols randomly, enabling a robust analysis of the feature accuracy under different noise conditions. After generating the symbols, we integrate them with signals received from 40 concurrently transmitting nodes and extract the CFO, STO, amplitude, and phase information from both antennas. For each data collection instance, we randomly select a starting chip and retrieve the subsequent 2^{sf} chips (representing a chirp signal), which is then combined with a randomly scaled target symbol. If the target symbol encounters a frequency tone collision—the frequency tone that encodes the symbol is within five bins of other concurrent symbols—we discard the chirp signal and attempt a new one.

Fig. 5 illustrates the estimation errors for CFO, STO, compensated amplitude, and phase across high, medium, and low SNR symbols. The results reveal that the average estimation error for CFO and STO across High, Medium, and Low SNR packets ranges from 0.7% to 2.0%, while the average estimation error for amplitude and phase remains between 3.0% and 9.7%. These findings indicate that the features of symbols across different SNR conditions can be accurately estimated.

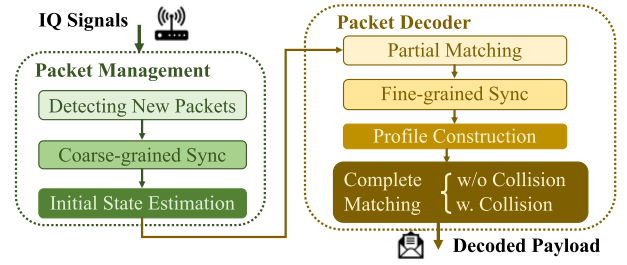


Fig. 6. Design architecture of Hi²LoRa.

2) *Collision Suppression With Two Receiving Antennas:* We conducted a simulation-based experiment to investigate the probability of frequency tone collisions under varying levels of network concurrency. In this experiment, the number of concurrently transmitting nodes ranges from 10 to 40. The packet length for each node is set to 50 symbols, and the Spreading Factor (SF) is fixed at SF10.

To test worst-case performance, each node operates with a near 100% duty cycle; that is, after transmitting a packet, each node randomly waits for one to two symbols before starting the transmission of another packet. A symbol is considered to experience a frequency tone collision if the frequency tone that carries the symbol is within five frequency bins of another symbol.

The collision probability, demonstrated in Fig. 1(c), reveals that over 23.7% of symbols are affected by frequency tone collisions. This issue becomes even more pronounced for low signal-to-noise ratio (SNR) packets. Conventional approaches, on the other hand, neglect the frequency tone collisions issue, as they primarily cater to networks characterized by low concurrency, in which frequency tone collisions merely occur.

Fortunately, recent advancements in multiple antennas allow concurrent packets to be separated using spatial multiplexing, significantly improving system performance [15], [16]. While frequency tone collisions are frequent, most collisions occur only between two frequency tones, as shown in Fig. 1(c). For example, when there are 40 concurrent transmitting nodes, the probability of three or more symbols colliding with each other is no more than 5.5%. This makes it reasonable for us to use two receiving antennas to suppress the strongest interference from collided symbols.

Though the simulation is intended for illustrative purposes. While the simulation setting is relatively simple, it demonstrates the trend that frequency collisions become more frequent as network concurrency increases. Additionally, the results suggest that frequency tone collisions occur most often between two concurrently transmitting nodes.

III. DESIGN OVERVIEW AND CHALLENGES

A. Design in a Nutshell

Fig. 6 illustrates the design architecture of Hi²LoRa.

The packet management module detects and records the arrival and departure of a packet. When a packet arrives, it performs

coarse-grained synchronization between the transmitter and receiver pair. Notice that, achieving chip-level synchronization is challenging at this stage, particularly as down chirps may face interference from concurrent transmissions. The initial states of the packet, such as its initial CFO (Carrier Frequency Offsets), STO (Sampling Time Offsets) as well as the initial channel states are extracted from the preambles and sync words of the newly arrived packet.

Once the information in the preambles are processed, the receiver passes the remaining signals to the packet decoder. The packet decoder performs two matching rounds to derive the final payload. First, it decodes a subset of symbols in a packet. Symbols can be decoded successfully in the first round if there is one and only one frequency tone that can be matched to the symbol. The first round matching only leverages the offsets information, thus does not require accurate chip-level synchronization between the transceiver.

After the partial matching module, a subset of symbols will be decoded. The offsets and phase information of these symbols will be used to align all chips in a symbol with the target packet, thus achieve chip-level synchronization between the transceiver. The fine-grained synchronization procedure is crucial to extract the packet profile with high accuracy as the chip level misalignment introduces high phase variations of a frequency tone. If not handled properly, such phase variations will lead to inaccurate channel estimation, and imprecise profile construction.

The complete profile of symbols yet to be decoded is constructed using the profile construction module after the receiver synchronizes to the transmitter finely. This sub-module input the offsets and channel state information of the already decoded symbols into the specially designed BiLSTM networks to estimate the profile of the symbols remain to be decoded.

In the second round, the decoder adopts the complete matching module to decode the rest of the symbols. In case frequency tone collision occurs, it exploits signals from two receiving antennas to suppress the collisions before examining the features. All the symbols will be decoded after the complete matching module.

B. Design Challenges

1) *How to Extract the Initial State of a Packet in Highly Concurrent Networks:* Extracting the initial state accurately is crucial for decoding a packet. However, a large number of concurrent packets will bring in excessive interference to each other, bringing in imprecise state estimation if symbol are evaluated separately. We introduce a novel estimation approach that extracts the initial states over all the preambles and the sync words jointly. The details are provided in Section IV-C.

2) *How to Construct a Packet's Time-Varying Profile:* Since the precision of packet profiles is essential in identifying and disentangling concurrent packets, algorithms should be carefully designed to trace the time variations of the profiles. We introduce a fine-grained synchronization algorithm to facilitate accurate channel state tracking in Section V-B. Moreover, a Bi-LSTM network structure is proposed in Section V-C for predicting the feature profile of a packet.

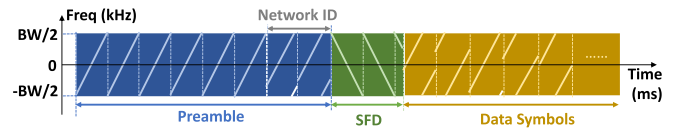


Fig. 7. The PHY-layer LoRa Packet format.

3) *How to Suppress Frequency Tone Collisions With Low Computation:* An efficient yet easy to implement approach is required to confront with the signal corruption caused by other colliding frequency tones (from different transmitters). This challenge is detailed in Section V-E.

IV. DESIGN OF THE PACKET MANAGEMENT

This section introduces the design of the Packet Management module. This module first detects the existence of a newly arrived packet, synchronizes to the new packet, and extract the initial states in the packet.

A. Detecting Newly Arrived Packets

The PHY-layer LoRa packet format is illustrated in Fig. 7. A LoRa packet starts with a set of preamble chirps (usually 6 to 8 preamble symbols). The former part of the preamble are basic up-chirp symbols, and the last two preamble symbols carry the network ID associated with a packet. 2.25 down-chirp symbols (called Start Frame Delimiter, SFD) follows the preamble. The SFD is used for synchronizing a LoRa packet in standard LoRa PHY. Data symbols are then organized subsequently.

When network concurrency is low, the receiver can easily detect and synchronize with a packet by checking the boundary of its SFD. However, since the up-chirping signals (symbols from other transmitters) are not strictly orthogonal to the down-chirping SFD, the more on-going transmissions, the higher interference to the SFD, and the more difficult to identify the SFD correctly, making it impractical for synchronizing concurrent packets in high-concurrency networks.

We thus bypass the use of SFD for packet detection and synchronization. Instead, we rely on preamble chirps for detection and coarse-grained synchronization. The detection process begins with a sliding window initialized to a length of 6 chirps (the minimum number of preamble symbols in LoRa packets [17]). The window slides $\frac{m}{8}$ chips if no packet is detected. Otherwise, it will enter the coarse-grained packet synchronization module, in which the window size will be adjusted to encompass all the preamble chirps.

We first discuss how to detect the presence of a new packet with a given detection window. The packet detection module computes the amplitude of each chirp within this window. Since the phase information of the newly arrived packet is unavailable, the amplitude of the preamble chirps is approximated by summing the absolute values of the frequency bins from two antennas. Let $u_{s,v,1}$ and $u_{s,v,2}$ represent the amplitude of the s th chirp at frequency tone v from antennas 1 and 2, respectively. The overall amplitude of frequency tone v from both antennas is approximated as $|u_{s,v,1}| + |u_{s,v,2}|$. When a new packet arrives,

the overall amplitude of a frequency tone v remains to be high in consecutive chirps.

Besides the base up-chirps, the last two chirps in the preamble carry the network ID, which are known values. Let the transmitted symbols of these last two preamble chirps be denoted as $v_{ID,1}$ and $v_{ID,2}$, respectively. A newly arrived packet is detected (with base up-chirp carried by frequency tone v) if the following conditions are met:

- 1) There exists a frequency tone v , and the amplitude of frequency tone v of at least two preamble chirps before the network ID exceed the threshold;
- 2) The amplitude of the frequency tone $v_{ID,1} + v$ of the penultimate chirp exceeds the threshold; or the amplitude of the frequency tone $v_{ID,2} + v$ of the final chirp exceeds the threshold; or both amplitudes exceed the threshold.

B. Coarse-Grained Packet Synchronization

Once detecting a new packet, the detection window is adjusted to encompass all the preamble chirps, thus synchronize the packet with the receiver. This procedure consists of two part: a. searching for all the base up-chirp symbols; and b. coarsely synchronizing to the packet.

We first examined the chirps prior to the detection window one by one. For the currently inspected chirp, if the amplitude of its b th frequency tone also exceeds the threshold, the chirp will be included into the detection window. The detection window expands until no base up-chirp symbol can be observed outside the detection window.

Once detecting all possible base up-chirps, the detection window slides chip by chip, and the accumulated amplitude of all the frequency bins of the base up-chirps and the network ID chirps are calculated. The starting point of the detection window who has the largest cumulative amplitude is determined as the starting point of the new packet. Up to this point, the coarse-grained synchronization can be achieved.

Once obtaining the detection window, it searches for the starting chip of the first preamble symbol in the window with the highest cumulative amplitude.

Currently, only coarse-grained synchronization is feasible due to the unavailability of accurate synchronization using SFD. This limitation is exemplified in the Fine-grained synchronization sub-module to be discussed in Section V-B.

Please be aware that though matching all the preamble chirps can effectively detect a newly arrived packet with low SNR, it is still not able to resolve concurrent packets with overlapping preambles, i.e., two packets whose preamble chirps experience frequency tone collisions. However, the probability that preamble chirps from two concurrent packets suffer from frequency tone collisions is extremely low for networks with medium concurrency (i.e., 40 nodes with SF=10). Therefore, this situation will not cause significant performance degradation.

C. Initial States Estimation

The last sub-module in the Packet Detector strives to estimate the initial fine-grained states of the newly detected

TABLE I
NOTATIONS

m	The number of chips in a symbol.
z_s	The value of the s th symbol transmitted in a packet.
$x_{s,i}$	The i th received chip of the s th chirp.
$w_{s,i}$	The noise from the i th received chip of the s th chirp.
τ_s	The overall STO of the s th symbol in a packet.
δ_s	The CFO of the s th symbol.
ζ_s	The SFO of the s th symbol.
f_s	The overall frequency drift due to CFO and STO.
$\phi_{s,k}$	The phase of the frequency tone carries the s th symbol from antenna k , $k \in \{1, 2\}$.
$\phi_{\text{CFO},s}$	The phase drift at the s th symbol due to CFO.
$\phi_{\text{STO},s}$	The phase drift at the s th symbol due to STO.
Δ_s	The phase difference from two receiving antennas of the frequency tone carries the s th symbol
$a_{s,k}$	The amplitude of the frequency tone carries the s th symbol from antenna k .
$a'_{s,k,v}$	The amplitude of the s th chirp from antenna k evaluated with frequency tone v .
$h_{s,k}$	The channel state of the s th symbol, and the k th antenna.
$\delta'_{s,k,v}$	The CFO of the s th chirp from antenna k evaluated with frequency tone v .
$\tau'_{s,k,v}$	The STO of the s th chirp from antenna k evaluated with frequency tone v .
$\phi'_{s,k,v}$	The phase of the s th chirp from antenna k evaluated with frequency tone v .

packet before decoding it. In particular, the set of initial states to be extracted are the fine-grained CFO, STO, as well as the average amplitude of the preamble chirps and the phase difference of the preambles chirps from the two receiving antennas.

1) *Retrieving Fine-Grained Offsets From the Given Signal:* We first demonstrate the impact of the offsets on the received signal. Table I lists commonly used notes in the current and the subsequent sections.

In Chirp Spread Spectrum (CSS), a symbol (also known as a chirp) is composed of $m = 2^{sf}$ chips, and sf is a parameter adaptive to a receiver's SNR.

Since a transceiver pair uses separate oscillators to transmit and received data, several offsets, such as CFO (Carrier Frequency Offset), STO (Sampling Time Offset) and SFO (Sampling Frequency Offset), exist between the transmitter and the receiver. CFO introduces an additional frequency drift at the received signal; STO, as will be shown in (1), causes a phase drifts between chips in a symbol; SFO, changes the STO over time, introducing time variations to the features used for parallel decoding.

To fully explores the features, Hi²LoRa, thus, leverages a general approach to extract CFO, STO and SFO jointly with high precision. For brevity, we only demonstrate the offsets of one antenna. The derivation of signals from multiple antennas can be extended directly from the discussion. In particular, let ζ_s , δ_s , and τ_s be the SFO, CFO and STO of the s th symbol observed by the gateway. Similar to the approaches discussed in CONST [18] and Magnifier [12], the de-chirped signal at one of the receiving antenna (denoted as $x_{s,i}$) can be approximated as

$$x_{s,i} = \begin{cases} h_s e^{j2\pi\phi_s} e^{j2\pi(z_s + f_s)\frac{i}{m}} + w_{s,i}, & 0 \leq i < m - z_s, \\ h_s e^{j2\pi\hat{\phi}_s} e^{j2\pi(z_s + f_s)\frac{i}{m}} + w_{s,i}, & m - z_s \leq i < m. \end{cases} \quad (1)$$

Here z_s is the value of the s th symbol, h_s is the channel state, and $w_{s,i}$ is the environmental noise observed at the i th chip of symbol s after the de-chirping operation. The phase of the signal is normalized with 2π , enabling a clearer representation of the relationship between offset-induced phase and frequency drift as functions of CFO and STO, as will be discussed in the rest of the section.

With offsets, the i -th transmitted chip in the s -th chirp will be received and sampled as the $(i + \tau_s)$ -th chip, where τ_s is the STO of the s -th chirp. In other words, the $(i - \tau_s)$ -th transmitted chip will be sampled and processed as the i -th chip at the receiver. By substituting $(i - \tau_s)$ with i in (1), demodulating it with the standard de-chirping and FFT operation, we can derive (2a-2h).

Denote $\phi_s = \phi_{\text{CFO},s} + \phi_{\text{STO},s}$; $\phi_{\text{CFO},s}$ and $\phi_{\text{STO},s}$ as the cumulative phase **changes due to CFO and STO up to the s -th chirp**. For instance, when $s = 1$, $\phi_{\text{CFO},s} = 0$; and when $s = 2$, $\phi_{\text{CFO},s} = \delta_1$. Also let f_s be the overall frequency drift of the s th symbol, and ϕ_s be the overall phase of a received chirp. These value can be expressed as:

$$\phi_{\text{CFO},s} = \sum_{1 \leq i \leq s-1} \delta_i, \quad (2a)$$

$$\phi_{\text{STO},s} = -\frac{z_s \tau_s}{m} + \frac{\tau_s}{2} + \frac{\tau_s^2}{2m}, \quad (2b)$$

$$\hat{\phi}_{\text{STO},s} = \phi_{\text{STO},s} + \tau_s, \quad (2c)$$

$$f_s \approx -\tau_s + \delta_s, \quad (2d)$$

$$\tau_s \approx \tau_0 + \sum_{1 \leq i \leq s-1} (m-1)\zeta_i, \quad (2e)$$

$$a_s = \left| \sum_{0 \leq i < m-z_s} x_{s,i} e^{-j2\pi(z_s + f_s)\frac{i}{m}} + e^{-j2\pi\tau_s} \left(\sum_{m-z_s \leq i < m} x_{s,i} e^{-j2\pi(z_s + f_s)\frac{i}{m}} \right) \right| \quad (2f)$$

$$\approx m|h_s| + O(\sqrt{m\gamma}) \approx m|h_s|. \quad (2g)$$

$$\phi_s = \phi_{\text{CFO},s} + \phi_{\text{STO},s} + \frac{\angle h_s}{2\pi} \quad (2h)$$

Here γ is the energy of the environmental noise, and $\angle h_s$ represents the angle of the channel state h_s . Even for a low Signal to Noise Ratio (SNR) packet, the noise term $O(\sqrt{m\gamma})$ is much lower than the received signal $m|h_s|$. For instance, when SNR = -8 dB for SF10 packets, $m = 2^{10} = 1024$, and $O(\sqrt{m\gamma}) \sim O(\frac{m|h_s|}{40}) \ll m|h_s|$. Therefore, we can omit the noise term in the following calculation. Note that, in (2), we only consider the impact of hardware imperfections on the received signal, without accounting for frequency tone collisions from other nodes. In this context, the interference from concurrent packets can also be approximated as noise. The case of symbols experiencing frequency tone collisions will be discussed in Section V-E. Additionally, as SNR keeps decreasing, the noise term can distort the accuracy of the features. However, this is a common problem for parallel decoding, but not a unique issue for our approach. We leave this issue as our future work. In

addition, (2h) is the amplitude of a symbol after compensating for offsets. As shown in Fig. 4(b), the compensated amplitude achieves higher processing gain, resulting in more stable feature estimations. We employ Zoom-FFT [19], and the approaches in Magnifier [12] to search for the frequency drift f_s and calculate the STO τ_s .

Please be aware that in this paper, the fine-grained offsets of signals received by two antennas are assume to be the same. The reason are the following. First, the receiver gets two different signal copies from two antennas of the same gateway. As the reception of two antennas is controlled by the same oscillator, the CFO and SFO of the two copies are almost the same. Second, since the distance between two antennas is only several centimeters to several tens of centimeters, the STO observed by different antennas are extremely close. For instance, denote the distance between the two antennas as d_a , the speed of light as c , the bandwidth of the signal as W , the STO difference between two antenna is $\frac{d_a W}{c} = 2.5 * 10^{-5}$ chip if $d_a = 0.3$ m and $W = 250$ KHz. We thus can safely ignore the offsets different between two antennas.

2) *Estimating the Initial States of the Packet*: The previous subsection discusses how to retrieve fine grained offsets information within a chirp. Due to collisions and noise, the estimated offsets can deviate from the actual value. In addition, accurate initial state estimation is crucial to decode a packet successfully as the features in a packet change over time, and we can only leverage the preceding and current features to estimate subsequent ones. In this subsection, we introduce an algorithm to estimate the fine-grained offsets of all the symbols in preamble jointly. Notably, the joint estimation approach aims to provide a more accurate initial state (i.e., the initial CFO, STO, amplitude and phase) estimation for decoding a packet.

Denote Δ'_s as the phase difference between signals from antenna 1 and 2 of **preamble s** , as there are only six to eight preamble chirps (24 ms to 32 ms for SF9 packets with bandwidth = 125 KHz), we **assume the Channel State Information (CSI) remain stable when transmitting the preamble chirps, and approximate Δ'_s to a fixed value Δ_0** . In addition, **CFO and SFO are assumed to remain stable for all preamble symbols** to ease the modeling of the phase drift due to CFO ($\phi_{\text{CFO},s}$) and STO ($\phi_{\text{STO},s}$). With this, CFO (δ_s) and SFO (ζ_s) is replaced with the fixed initial value δ_0 and ζ_0 that do not change across symbols. Ideally, phase and frequency drifts in preamble chirps without frequency tone collisions are

$$\phi_{\text{CFO},s} \approx \phi_{\text{CFO},s-1} + \delta_0, \quad (3a)$$

$$\phi_{\text{STO},s} \approx \phi_{\text{STO},s-1} + \frac{m(\zeta_0 - 1)}{2}, \quad (3b)$$

$$\tau_s \approx \tau_{s-1} + m(\zeta_0 - 1) \quad (3c)$$

$$\phi_s \approx \phi_{s-1} + \delta_0 + \frac{m(\zeta_0 - 1)}{2}, \quad (3d)$$

$$f_s \approx f_{s-1} + m(\zeta_0 - 1). \quad (3e)$$

Here (3a) and (3b) is derived from (2a) and (2b), respectively; and the fact that preambles correspond to symbol 0 ($z_s = 0$) and $\tau_s \ll m$. (3c) is feasible as the STO changes is mainly caused by the SFO between the transceiver pair. (3d)–(3e) are

derived from (2d) and (2h). Let τ' be the STO of the first detected preamble. With (3), the phase (ϕ_s) and frequency drift (f_s) becomes

$$\phi_s \approx \phi_0 + s\delta_0 + \frac{\tau' + sm(\zeta_0 - 1)}{2}, \quad (4a)$$

$$f_s \approx \delta_0 + \tau' + sm(\zeta_0 - 1). \quad (4b)$$

We perform regressions on the phase and frequency of all the preambles with (4) to derive Δ_0 , δ_0 and ζ_0 . We have:

$$\arg \max_{\{\Delta_0, \tau', \delta_0, \zeta_0\}} \left| \sum_s e^{-j2\pi\phi_s} (a'_{s,1,f_s} + e^{j2\pi\Delta_0} a'_{s,2,f_s}) \right| \quad (5a)$$

$$\text{s.t.} \quad (2f), (4a), (4b) \quad (5b)$$

where $a'_{s,1,f_s}$ is the amplitude of the s th preamble chirp evaluated with frequency drift f_s (refer to (2f)). Note that the preamble chirps are transmitted with symbol 0, and received with frequency tone f_s (due to offsets). By multiplying $e^{j2\pi\Delta_0}$ to the signals from the second antenna, we compensate for the phase difference between two antennas, thus use the overall amplitude for initial state estimation.

The initial states are consisted of the estimated CFO, STO of the first data symbol, the phase difference between the signal from two receiving antennas, and the estimated amplitude of the first data symbol from the two receiving antennas. Suppose there are n_s preamble chirps, with (5), we can retrieve the initial CFO and the phase difference as δ_0 and Δ_0 , respectively. The estimated STO (denoted by τ_0) and the estimated amplitude from two antennas becomes

$$\tau_0 = \tau' + (n_s + 2.25)m(\zeta_0 - 1) \quad (6a)$$

$$a_{0,k} = \frac{1}{n_s} \sum_s |a'_{s,k,f_s}|, \quad k \in \{0, 1\}. \quad (6b)$$

Here the estimated initial STO of the first data symbol is calculated by considering the accumulated sampling timing drift of n_s preambles and 2.25 down-chirp SFD symbols. τ_0 is calculated with the accumulated sampling time offsets after $(n_s + 2.25)m$ chips. In total, the initial states to be estimated are $\{\delta_0, \tau_0, \Delta_0, a_{0,1}, a_{0,2}\}$.

Note that, as will be discussed, the initial states will be used for partial matching module. Therefore, we do not need to calculate the phase of the first data symbol from each antenna, but only need the phase difference information.

V. DESIGN OF THE PACKET DECODER

This section provides all the necessary details to decode a concurrently transmitted packet.

A. Partial Matching

The Packet Decoder first leverages the limited features, e.g., the CFO and STO information for the Partial Matching sub-module.

In particular, this sub-module decodes a subset of symbols using only offset features. The process involves two steps: first,

selecting a set of frequency tone candidates through frequency tone pre-screening, and second, matching one frequency tone from the pre-screening set and decoding it as the target symbol. It's important to note that this module performs matching for a set of symbols in a packet based solely on offset features. There is a possibility that some symbols may not find a match with any frequency tone due to the limited number of features. Any unmatched symbols will be addressed in subsequent modules.

The frequency tones of demodulated symbols are usually surrounded by side-lobes due to various offsets and arrival time misalignment among received packets. This module leverages a frequency bin pre-screening procedure to select the true frequency tones from the side-lobes when decoding symbols. Specifically, it uses a standard FFT procedure (as done in a standard LoRa decoder) to evaluate the spectrum distribution of each integer frequency bin. We then search for a set of frequency bins, and inspect their fine-grained offset features only if the frequency bins are local maximal. E.g., frequency bin v is a local maximal if $a'_{s,k,v} > a'_{s,k,v-1}$ and $a'_{s,k,v} > a'_{s,k,v+1}$, where $a'_{s,k,v}$ is the amplitude of frequency bin v of the s th symbol from antenna k . The pre-screening procedure can reduce the number of frequency tones to be evaluated, and avoid decoding ambiguity caused by side-lobes.

Denote the selected frequency tones from the pre-screening procedure as V_s ; the estimated CFO and STO of the s th symbol in the packet as δ_s and τ_s respectively; the CFO and STO evaluated at frequency tone v from antenna k as $\tau'_{s,k,v}$ and $\delta'_{s,k,v}$. For any frequency tone $v \in V_s$, we evaluate its STO and CFO from two antennas with (2b)–(2d). If there is only one frequency tone v that meets the criteria listed in (7), v will be matched to symbol s and $\tilde{\delta}_s$ and $\tilde{\tau}_s$ will be updated as

$$|\delta'_{s,k,v} - \delta_s| \leq \Gamma_\delta, \quad k \in \{1, 2\} \quad (7a)$$

$$|\tau'_{s,k,v} - \tau_s| \leq \Gamma_\tau, \quad k \in \{1, 2\} \quad (7b)$$

$$\delta'_{s,v} = \frac{a'_{s,1,v}\delta'_{s,1,v} + a'_{s,2,v}\delta'_{s,2,v}}{a_{s,1,v} + a'_{s,2,v}}, \quad (7c)$$

$$\tau'_{s,v} = \frac{a'_{s,1,v}\tau'_{s,1,v} + a'_{s,2,v}\tau'_{s,2,v}}{a_{s,1,v} + a'_{s,2,v}}, \quad (7d)$$

$$\delta_{s+1} = (1 - \rho_\delta)\delta_s + \rho_\delta\delta'_{s,v}, \quad (7e)$$

$$\tau_{s+1} = (1 - \rho_\tau)\tau_s + \rho_\tau\tau'_{s,v}. \quad (7f)$$

Here $a'_{s,k,v}$ is the amplitude of frequency tone v from antenna k . (7c)–(7d) calculate its average offsets. $\delta'_{s,v}$ and $\tau'_{s,v}$ are then fed into the low pass filter (7e)–(7f) to smooth the estimation. ρ_δ and ρ_τ are two parameters controlling the changing rate of CFO and STO. If no frequency tone is paired with the s th symbol, $\delta_s = \delta_{s-1}$ and $\tau_s = \tau_{s-1}$. The intuition of (7) is that though the packet profile changes over time, adjacent symbols in a packet demonstrate similar profiles. Therefore, we can leverage this property to decode symbols that do not experience decoding ambiguity. (7c) and (7d) represent the average CFO and STO calculated from the two antennas. δ_{s+1} and τ_{s+1} denote the time-smoothed CFO and STO for the next symbol, respectively.

A low-pass filter is applied to smooth these values, ensuring stability and reducing noise.

B. Fine-Grained Chip Level Synchronization

This subsection details the methodology for achieving chip-level synchronization between the gateway and the transmitter. It is crucial to emphasize that chip-level synchronization forms the basis for precisely extracting the phases of each symbol. Let $\phi_{s,k}$ represent the phase information from antenna k after offset compensation. $\phi_{s,k}$ can be derived by deducing the CFO and STO induced offsets (e.g., (2a)–(2b)) from (2h). That is:

$$y_{s,k,1} = \sum_{0 \leq i \leq m-z_s-1} x_{s,i,k} e^{-j2\pi \frac{(z_s+f_s)i}{m}}, \quad (8a)$$

$$y_{s,k,2} = \sum_{m-z_s \leq i \leq m-1} x_{s,i,k} e^{-j2\pi \frac{(z_s+f_s)i}{m}}, \quad (8b)$$

$$\phi''_{s,k} = \frac{\angle(y_{s,k,1} + e^{-j2\pi \tau_s} y_{s,k,2})}{2\pi}, \quad (8c)$$

$$\phi_{s,k} = \phi''_{s,k} - \sum \delta_s + \frac{z_s \tau_s}{m} - \frac{\tau_s}{2} - \frac{\tau_s^2}{2m}, \quad (8d)$$

Here $x_{s,i,k}$ refers to the received i th chip of the s th chirp from the k th antenna. The notation $x_{s,i,k}$ is similar to $x_{s,i}$. While $x_{s,i}$ refers to the received i th chip of the s th chirp when there is only one antenna. According to (8d), it is evident that the phase of a symbol depends on the transmitting frequency tone. For instance, when frequency tone z_s is transmitted, it introduces a phase component $\frac{\tau_s z_s}{m}$. This value varies from -2π to 2π depending on the exact value of the transmitted symbol z_s and the STO τ_s between the transceiver. When $\frac{\tau_s z_s}{m}$ is not carefully compensated for in the phase of the frequency peak, it cause drastic variations even between adjacent symbols. Refer to Fig. 4(a), the phase feature is useless should phase compensation is performed.

Moreover, as the coarse-grained packet synchronization is not able to align the first chip of the symbol at the transmitter with the first chip received at the gateway, it is possible that STO and $z_s > 1$. Thus both integer and fractional part of τ_s should be retained in high precision so as to facilitate accurate phase tracking (e.g., $\phi_{s,k}$).

We next introduce an efficient strategy to finely align the receiver with the transmitter chip by chip. We first let S be a set of symbols decoded from the partial matching module, and $\Phi_S = \{\phi_s | k \in S\}$ be their phases. Also denote $\tau_s = \tau'_0 + \tau_{s,2}$ where τ'_0 refers to the initial integer STO, and $\tau_{s,2}$ is the estimated STO from (7). As $\frac{\tau_s^2}{2m} \ll 0$, the phase of each frequency peak after compensating for the offsets ($\phi_{s,k}$ from (8d)) can be simplified as

$$\phi_{s,k} \approx \phi''_{s,k} - \sum \delta_s + \left(\frac{z_s}{m} - \frac{1}{2}\right) \tau_s \quad (9a)$$

$$\approx \phi''_{s,k} - (s-1)\delta + \left(\frac{z_s}{m} - \frac{1}{2}\right) \tau_{s,2} + \left(\frac{z_s}{m} - \frac{1}{2}\right) \tau'_0. \quad (9b)$$

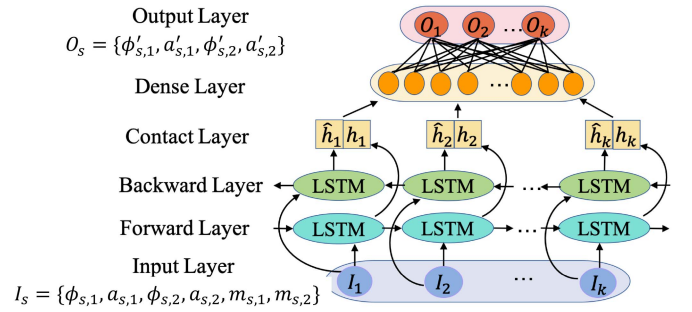


Fig. 8. The network structure of the proposed BiLSTM.

Here δ is the average CFO, and the compensated phase $\phi_{s,k}$ only carries the channel state experienced by symbol s . Moreover, channel state changes gradually, and only varies slightly for a small number of symbols. This implies that consecutive $\phi_{s,k}$ tends to be relatively stable too.

On the other hand, the last term $(\frac{z_s}{m} - \frac{1}{2})\tau'_0$ in (9) can results in significant changes of the compensated phase $\phi_{s,k}$. Based on the above observation, we conclude that the optimal STO, namely τ_0^* , should causes the least turbulence of the compensated phase information. Therefore, we use the Minimum Mean Square Error (MMSE) and the following optimization algorithm to search for the optimal τ_0^* .

$$\tau_0^* = \arg \min_{\tau'_0} \sum_{\{s \in S, s-1 \in S\}} |\phi_{s,k} - \phi_{s-1,k}|^2 \quad (10)$$

C. Profile Construction

Given the set of frequency tones selected from previous module, we construct the complete profile of the packet in this module.

1) *Workflow of the Profile Construction Sub-Module:* We first leverage a two layer BiLSTM networks to estimate the CFO and STO for symbols yet to be decoded. Once the per symbol STO and CFO are available, they are utilized to compensate for the phase and amplitude of the actual frequency peak of all the already matched symbols with (2). The amplitude and phase profile of the already matched symbols are then fed into another BiLSTM network to construct the complete profile of a packet.

Notably, two separate Bi-LSTM network are leveraged for constructing the packet profile. The first one is for CFO and STO estimation, and the second one is for amplitude and phase estimation. We use the network for amplitude and phase estimation (channel state estimation) as an example to illustrate how the Bi-LSTM is constructed.

2) *Network Structure of the Bi-LSTM Network:* The architecture of the BiLSTM network is plotted in Fig. 8. When constructing packet profiles, each symbol within the packet is associated with a pair of masks, denoted as $\{m_{s,1}, m_{s,2}\}$. $m_{s,1}=1$ if the s th symbol is decoded in the partial matching sub-module, and 0 otherwise. Given $m_{s,1}=0$, quadratic regression is utilized to make the initial predictions for the unmatched symbols. The second mask $m_{s,2}$ indicates the length of a packet. That is

$m_{s,2} = 1$ if the packet length is no smaller than s , and $m_{s,1} = 0$ otherwise.

The pair of mask, as well as the phase and amplitude information of symbol s from two antennas are organized into a vector $\{a_{s,1}, \phi_{s,1}, a_{s,2}, \phi_{s,2}, m_{s,1}, m_{s,2}\}$ as shown in Fig. 8. Moreover, the batch size is set to 100 data packets, and each packet lasts for 688 data symbols (the maximum payload size of a packet with any SF, i.e., a SF6 packet with CR 4/8 that contains 255 Bytes payload data). Batch Normalization is performed for the input data.

The Bi-LSTM network contains a forward and a backward LSTM layer, their output size is organized as (100, 688, 10). The contact layer concatenates the result from the two LSTM network as shown in Fig. 8, and passes them into the dense layer. The dense layer uses a Fully Connected (FC) network to make the final prediction. The output of the network is the estimated amplitude and the phase of all the symbols in a packet.

When training the network, 10000 packets with different channel conditions and packet length are collected. We then randomly drop 10% to 40% symbols in a packet to train the parameters. Data augmentation is performed by adding Gaussian noise to the received data. The loss function (denoted as L) is set as

$$L = \sum_{s,k \in \{1,2\}} |a_{s,k} e^{j2\pi\phi_{s,k}} - \tilde{a}_{s,k} e^{j2\pi\tilde{\phi}_{s,k}}|, \quad (11)$$

where $\tilde{a}_{s,k}$ and $\tilde{\phi}_{s,k}$ are the estimated amplitude and the phase information of the k antenna output by the network.

It is worth to note that CFO and STO estimation are relatively straightforward since CFO remains stable within a packet, and STO variation is primarily influenced by SFO, which is also stable within the packet duration. Traditional regression methods can therefore achieve high estimation accuracy. The Bi-LSTM is primarily designed for channel state estimation, which is more variable due to the longer duration of LoRa packets—often tens to hundreds of milliseconds—exceeding the wireless channel's coherence time. As the Bi-LSTM effectively handles both channel-sensitive and insensitive features, we leverage the same network for estimating offsets and channel states.

Overall, the Bi-LSTM network is capable of accurately predicting both hardware offsets, and the channel state variations over time, thereby significantly improves the accuracy of the estimated packet profile.

D. Complete Matching Without Frequency Tone Collisions

This module utilizes all available features, including both offsets and channel state information, to decode symbols that suffer from decoding ambiguity, but not frequency tone collisions. After obtaining the profile, all symbols will be matched with a frequency tone. Specifically, if a symbol does not experience frequency tone collision, we decode the symbol with a frequency tone whose offsets, amplitude, and phase features match the packet profile. If frequency tone collision occurs, we explore signals from two antennas to suppress signal corruption. In particular, let $\tilde{\delta}_s, \tilde{\tau}_s, \tilde{a}_{s,k}$ and $\tilde{\phi}_{s,k}$ be the values derived from the constructed profile, and $\{\tilde{\Gamma}_\delta, \tilde{\Gamma}_\tau, \tilde{\Gamma}_a, \tilde{\Gamma}_\phi\}$ are thresholds used for the matching. For symbol s yet to be decoded, we evaluate all

the features of a frequency tone $v \in V_s$ with (2) and (8), and choose the one that satisfies the conditions:

$$|\delta'_{s,k,v} - \tilde{\delta}_s| \leq \beta \tilde{\Gamma}_\delta, \quad k \in \{1, 2\} \quad (12a)$$

$$|\tau'_{s,k,v} - \tilde{\tau}_s| \leq \beta \tilde{\Gamma}_\tau, \quad k \in \{1, 2\} \quad (12b)$$

$$\frac{1}{1 + \beta \tilde{\Gamma}_a} \leq \frac{a'_{s,k,v}}{\tilde{a}_{s,k}} \leq 1 + \beta \tilde{\Gamma}_a, \quad k \in \{1, 2\} \quad (12c)$$

$$|\phi'_{s,k,v} - \tilde{\phi}_{s,k}| \leq \beta \tilde{\Gamma}_\phi. \quad k \in \{1, 2\} \quad (12d)$$

where $\phi'_{s,k,v}$ is the phase of the s th symbol evaluated at frequency tone v . Initially, β is set to 1. If there is only one frequency tone v^* satisfies (12), the symbol is decoded as v^* . If no frequency tone satisfies (12), it enters the frequency tone collision suppression sub-module shown later. If there are more than one frequency tones that satisfy (12), we choose the one that meets all the conditions in (12) with the least β . The exact values of the threshold parameters are discussed in Section V-F.

E. Complete Matching With Frequency Tone Collisions

We exploit spatial multiplexing with signals from two receiving antennas to suppress collisions. As has been illustrated in Section II, most symbols only experience frequency tone collision from one packet. E.g., with 40 concurrent packets, the probability that symbols suffer collisions from more than one packet is only 2.0% to 6.9%. Therefore, we aim at suppressing the most severe corruptions from one collided packet. In the following subsections, we first discuss how to suppress interference when the interfering node is known. Collision suppression from unknown interfering node is discussed later.

1) *Frequency Tone Collision Suppression With a Known Interferer c^** : Let the set of collided packets be C . Denote the estimated channel state of the target packet and the collided packets as $H_s = [h_{s,1}, h_{s,2}]^T$ and $H_c = [h_{c,1}, h_{c,2}]^T$, $c \in C$; the collided frequency component in the spectrum domain as z_t and z_c ; the received signals and the noise from two receiving antennas as $Q = [q_1, q_2]^T$ and $N_f = [n_1, n_2]^T$. Here $[\cdot]^T$ denotes the transpose operation of a matrix. Q becomes

$$Q = H_s z_s + \sum_{c \in C} H_c z_c + N. \quad (13)$$

Designate c^* as the packet that causes the strongest signal corruption to the target packet ($c^* \in C$), we construct a suppression vector of interferer c^* , denoted as W_{c^*} , where $W_{c^*} = [1, -h_{c^*,1}/h_{c^*,2}]$ and $W_{c^*} H_{c^*} = 0$; and a suppression validation vector for the s th symbol of the packet to be decoded as $W_s = [1, -h_{s,1}/h_{s,2}]$ and $W_s H_s = 0$. Now multiply Q with W_{c^*} and W_s respectively, we can get

$$\begin{aligned} W_{c^*} Q &= W_{c^*} H_s z_s + W_{c^*} \left(\sum_{c \in C} H_c z_c \right) + W_{c^*} N \\ &\approx W_{c^*} H_s z_s + W_{c^*} N, \end{aligned} \quad (14a)$$

$$\begin{aligned} W_s Q &= W_s \left(\sum_{c \in C} H_c z_c \right) + W_s N \approx W_s H_{c^*} z_{c^*} + W_s N. \end{aligned} \quad (14b)$$

Here the approximation is done by assuming there is one dominant interferer in the collided packet set C . Notably, to construct W_{c^*} for all possible interferers, we need to collect the Channel State Information (CSI) from all the transmitters. This can be easily achieved as a gateway is responsible to decode all the packets from all the transmitters. The gateway can thus facilitate CSI sharing when performing parallel decoding. In addition, since constructing W_{c^*} only requires retrieving the value of $h_{c^*,1}/h_{c^*,2}$, which can be easily retained once the complete profile of interferer c^* is constructed. That is $h_{c^*,1}/h_{c^*,2} \approx a'_{c^*,1}/a'_{c^*,2} e^{j2\pi(\phi'_{c^*,1}-\phi'_{c^*,2})}$, where $a'_{c^*,1}$, $a'_{c^*,2}$, $\phi'_{c^*,1}$ and $\phi'_{c^*,2}$ are the amplitude and phase of interfering node c^* estimated from the complete profile construction procedure. While the suppression vector W_{c^*} in (15a) mitigates the energy contribution from the dominant interference, the validation vector W_s in (15b) provides an additional layer of assurance by verifying that the dominant interference originates from the interferer c^* .

2) *Frequency Tone Collision Suppression From the Unknown Interfering Node*: If the interferer is known, collision can be easily resolved by applying the suppression vector W_{c^*} to the received signal. However, as the decoder has no prior knowledge about c^* , we need to search a frequency tone v , and an interferer c^* , such that, by applying W_{c^*} to v , the collided frequency tone carrying the symbol from the target packet (and the corresponding feature profile) can be well recovered.

Recall from (14a), by suppressing the collided packet, the amplitude, and phase information of the target packet from the two antennas will change accordingly. Hence we cannot utilize the packet profile before interference suppression for parallel decoding when encountering frequency tone collision. To this end, we construct a fine-grained effective profile for symbol s as $\{\tilde{\delta}_s, \tilde{\tau}_s, \tilde{a}_{s,c^*}, \tilde{\phi}_{s,c^*}\}$. In the fine-grained effective profile, $\tilde{\delta}_s$ and $\tilde{\tau}_s$ are the same as estimated by the complete profile constructed procedure as the linear operation in (14a) between the received signals will not change the fine-grained CFO and STO of a symbol. The amplitude (denoted as $\tilde{a}_{s,c^*}$) and phase (denoted as $\tilde{\phi}_{s,c^*}$) profiles are adjusted with $\tilde{a}_{s,c^*} = |W_{c^*} H_s|$, and $\tilde{\phi}_{s,c^*} = \angle(W_{c^*} H_s)$, where the operations $|\cdot|$ and $\angle(\cdot)$ take the amplitude and the phase of a complex value respectively. Naively, to enable parallel decoding with collision suppression, we need to construct the effective profile of all possible frequency tone i with all different c^* , and decode the symbol with the most probable frequency tone i . However, this process incurs intolerably high computation overhead. For instance, assessing the fine-grained features of a frequency bin demands $O(m^2)$ operations, where $m = 2^{sf}$ represents the number of chirps in a symbol. If m frequency bins need evaluation, the overall decoding complexity with frequency tone collision approaches to $O(m^3)$.

To address this issue, akin to the Partial Matching module detailed in Section V-A, we introduce a pre-screening method to eliminate a majority of frequency tones from performing interference suppression.

Specifically, let $D'_i = [d'_{i,1}, d'_{i,2}]$ where $d'_{i,k}$ is the amplitude of the i th frequency bin received from the k th antenna after the coarse-grained FFT operation. Note that, the coarse-grained FFT

operation is a standard operation applied to both standard LoRa PHY and Hi²LoRa (in the Partial Matching sub-module). Also denote $A'_s = [a'_{s,1}, a'_{s,2}]$ as the estimated fine-grained amplitude of the target frequency bin from two receiving antennas. A'_s is determined after running the Profile Construction sub-module.

Now, instead of evaluating the fine grained features directly, we first estimate the coarse-grained effective amplitude of various frequency tones with $W_c D'_i$, and the fine-grained effective amplitude with $W_c A'_s$. Frequency tone i will be excluded from further evaluation if the difference between its coarse-grained effective amplitude and the fine-grained effective amplitude exceeds a pre-defined threshold, that is

$$\frac{1}{\beta'} < \frac{|W_c D'_i|}{|W_c A'_s|} < \beta', \quad \forall 0 \leq i < 2^{sf}, c \in C. \quad (15)$$

Here β' is a parameter that are similar to those used in the complete matching (12) sub-module without frequency tone collisions. The likelihood of a noisy frequency bin satisfying (15) is low, e.g., say the probability is approximately 10%. This indicates that, on average, the condition in (15) effectively filters out 90% of noisy frequency bins, significantly reducing computational complexity during collision suppression of the received signals.

Initially, β' is set to one. If no frequency tone satisfies (15), β' is adjusted downward to the largest value that allows at least one frequency tone to meet the condition. Conversely, if multiple frequency bins satisfy (15), all of them are selected for further testing in subsequent steps.

Once a set of symbols that have passed the pre-screening process have been collected, the one satisfies all the following constraints will be selected

$$|\bar{\delta}_{s,v} - \tilde{\delta}_s| \leq \beta \tilde{\Gamma}_\delta, \quad (16a)$$

$$|\bar{\tau}_{s,v} - \tilde{\tau}_s| \leq \beta \tilde{\Gamma}_\tau, \quad (16b)$$

$$|\bar{\delta}_{v,c^*} - \tilde{\delta}_{c^*}| \leq \beta \tilde{\Gamma}_\delta, \quad (16c)$$

$$|\bar{\tau}_{v,c^*} - \tilde{\tau}_{c^*}| \leq \beta \tilde{\Gamma}_\tau, \quad (16d)$$

$$\frac{1}{1 + \beta \tilde{\Gamma}_a} \leq \frac{\bar{a}_{s,v}}{\tilde{a}_{s,c^*}} \leq 1 + \beta \tilde{\Gamma}_a, \quad (16e)$$

$$|\bar{\phi}_{s,v} - \tilde{\phi}_{s,c^*}| \leq \beta \tilde{\Gamma}_\phi. \quad (16f)$$

Here, $\bar{\delta}_{s,v}$ and $\bar{\tau}_{s,v}$, $\bar{a}_{s,v}$, and $\bar{\phi}_{s,v}$ are the effective CFO, STO, amplitude and phase information of the packet to be decoded (derived from (13a)). $\bar{\delta}_{v,c^*}$ and $\bar{\tau}_{v,c^*}$ are the effective CFO and STO for the interferer c^* derived with (13b). (16a)–(16b) and (16e)–(16f) evaluate whether the CFO, STO, effective amplitude, and effective phase of the frequency tone (post-interference suppression) align with the estimated effective profile. Meanwhile, (16c)–(16d) provide an additional verification to ensure that the dominant interference originates from the interferer c^* .

If there are multiple frequency tone that satisfy all the constraints in (16), one of them will be randomly chosen as the target one carried by the symbol. If no frequency tone satisfy the constraints, we decrease β to make one frequency tone satisfy the constraints.

While the proposed frequency tone collision suppression method effectively mitigates collisions caused by a single interferer, it is important to note that its effectiveness decreases when multiple interferers significantly obstruct the target symbol or when consecutive preambles from two packets experience frequency tone collisions. In such cases, the collision suppression capability of two antennas may be insufficient, and additional antennas become necessary. However, the likelihood of these scenarios is low (as shown in Fig. 1(c), the probability of three or more symbols using the same frequency tone is below 5.5). As a result, two antennas are generally sufficient to handle the majority of frequency tone collisions.

3) *Incorporating Collision Suppression Into Hi²LoRa*: To put everything together, the frequency tone collision suppression procedure works as follows.

First, it uses a limited number of features to identify symbols without decoding ambiguity, as explained in Section V-A, “Partial Matching” (page 7). A symbol can be decoded in the “Partial Matching” module if there is exactly one frequency bin whose CFO and STO match the estimated CFO and STO of the packet. Once a set of unambiguous symbols is decoded, the phase and amplitude of these symbols are computed and treated as a subset of the channel state for the packet. This subset is then input into a Bi-LSTM, which estimates the channel state for the entire packet.

The decoder leverages the estimated channel state to suppress frequency tone collisions. As illustrated in Fig 9 and 10 (page 12), the channel state estimation error remains below 15%, even for low-SNR packets, when only 40% of the symbols are decoded via the “Partial Matching” module. This two-stage decoding process, combined with the Bi-LSTM approach, provides a robust mechanism for mitigating frequency tone interference.

Additionally, interference suppression is employed only when the gateway is unable to decode the packet during the “Full Matching” procedure, as discussed in Section V-D. The rationale is as follows: Since fine-grained offsets are fully compensated before the full matching process, the inability to match a symbol to any frequency bin suggests a strong collision has occurred, resulting in significant corruption of the features carried by the frequency bin. In such scenarios, spatial multiplexing is utilized to resolve the collision and facilitate the decoding of the symbol.

F. Parameter Selections

We next show how to determine the exact value of the parameters discussed in both partial and complete matching. Different packets with different SNR experience different degree of collisions and decoding ambiguities. Therefore, they should be assigned with different thresholds (i.e., Γ_δ , Γ_τ) and update parameters (i.e., ρ_δ , and ρ_τ). We categorize a SF10 packet as a low, medium or high SNR packet if its average SNR satisfies ≤ 0 dB, 0 dB $\leq$ SNR ≤ 10 dB, and ≥ 10 dB. For SF8 packets, we add a 6 dB additional SNR margin when categorizing SF8 packets as they have 6 dB less processing gain than SF10 packets. In other words, a SF8 packet with SNR > 16 dB, or SNR < -4 dB is assumed to be a high or low SNR packet.

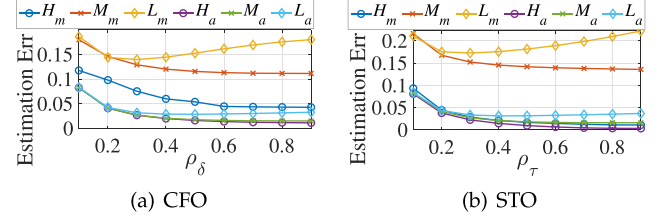


Fig. 9. H_m , M_m and L_m represent the 95th percentile of the CFO and STO estimation errors for packets with SNR = 10 dB, 0 dB and -10 dB respectively. H_a , M_a , and L_a represent average estimation errors.

TABLE II
PARAMETERS

	ρ_δ	ρ_τ	Γ_δ	Γ_τ	$\bar{\Gamma}_\delta$	$\bar{\Gamma}_\tau$	$\bar{\Gamma}_a$	$\bar{\Gamma}_\phi$
High (SNR)	0.9	0.9	0.05	0.05	0.08	0.08	0.08	0.08
Medium	0.6	0.6	0.15	0.15	0.1	0.1	0.12	0.12
Low	0.3	0.3	0.2	0.2	0.15	0.15	0.15	0.15

Specifically, a small ρ_δ suppresses noise better, but takes longer time to adapt to the time-varying offsets. Thus, a high SNR packet requires a higher ρ_δ , while a low SNR packet shows better performance with a lower ρ_δ .

To study the most appropriate value of ρ_τ , we collect ten SF10 packets from ten different nodes with an average SNR = 50 dB. The offset values from the packets are used as benchmarks. Gaussian noise is manually added to the packets to manipulate their SNR. According to Fig. 15 in the evaluations, the partial matching procedure can successfully matches 40% or more symbols. Therefore, we randomly select 40% of the symbols, and use them to predict the offsets of the rest symbols with the low pass filters introduced in Section V-A. Fig. 9 plots the average and the 95th percentile CFO and STO estimation error for packets with SNR = 10 dB, 0 dB and -10 dB respectively.

Based on the figure, the optimal ρ_δ and ρ_τ are determined as the ones with the lowest average estimation errors. Moreover, Γ_δ and Γ_τ are set to tolerate the 95th percentile estimation error. In this case, a collision free symbol can be correctly identified by the offsets (STO and CFO) from both antennas with probability higher than 80% ($\approx 0.95^4$, with dimension size = 4, which includes CFO and STO from two antennas). Their values are shown in Table II.

Regarding the Bi-LSTM network, an initial dataset of 1000 packets is compiled from diverse locations, and their channel state is calculated symbol by symbol. Data augmentation is applied to create additional packets with varying noise levels. Specifically, 60% of symbols are randomly chosen and labeled as unpaired symbols. The channel state of the remaining symbols is then fed into the Bi-LSTM network to train its parameters.

Fig. 10(a) demonstrates the average and the maximum amplitude and phase deviation when 60% of the channel state information is missing. Note that, in real evaluations (refer to Fig. 15 in the evaluations, Section V), partial matching module is able to pair more than 40% of symbols. We therefore, set the average deviation with 60% symbol loss as the threshold for the complete matching in Section V-D. Similarly, the average and maximum deviation for CFO and STO when 60% of the symbols

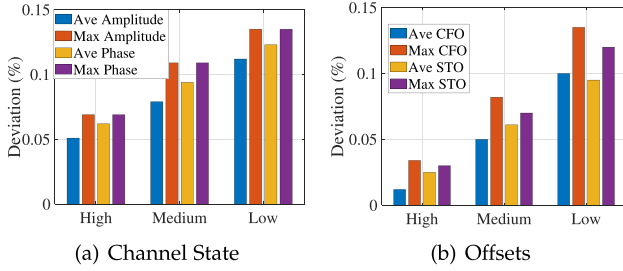


Fig. 10. (a). The average and maximum amplitude-phase deviation of the estimated data from the real ones. (b). The average and maximum offsets deviation from the estimated data.

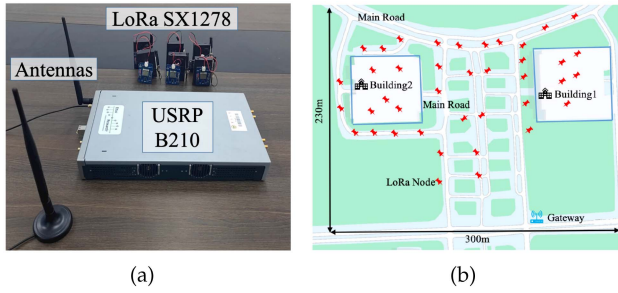


Fig. 11. (a) Devices used for the evaluations. (b). Experimental deployment of a 40 node network. The map size is 300 m $\times$ 230 m. The red pins represent LoRa transmitters, and the blue gateway is the receiver.

in a packet are missing are shown in Fig. 10(b). The values for each parameter are listed in Table II.

Moreover, in the Complete Matching sub-module, as every symbol should be matched with one and only one frequency tone, a fixed threshold can not satisfy this condition all the time. We therefore, let the parameter β , β' and $\tilde{\Gamma}'_a$ be adjustable. In particular, we set the initial value of β' and β' be one, and the initial value of $\tilde{\Gamma}'_a$ to $\tilde{\Gamma}_a$. If no frequency tone is selected or more than one frequency tone satisfies the above conditions, we adjust the threshold as discussed previously.

VI. EVALUATIONS

A. Experimental Settings

Commodity LoRa chips SX1278 [20] are used as the transmitters. A USRP B210 equipped with two antennas is employed as a LoRa gateway to decode concurrent packets. The hardware testbed is shown in Fig. 11(a). A computer is connected to the USRP to collect raw signals.

For each transmitter, the transmission power is set to 0 dBm. The bandwidth is 250 KHz. The central frequency band used for transmitting data is 510 MHz. In this case, two antennas at the receiver are placed 0.3 meters apart so that the antenna separation equals half of the wavelength. In each round of evaluation, all the nodes transmit data on the same channel and with the same SF. Network performance with different SF (SF8 and SF10) is investigated in different evaluation rounds. The Coding Rate (CR) is $\frac{4}{8}$. Notably, with CR4/8, one bit error can be recovered

for every received block of bits. The preamble length is eight symbols. Two Sync words and 2.25 SFD follow the preamble.

According to [20], with the above setting, the packet length is 46.25 and 38.25 symbols for SF8 and SF10 respectively. The packet duration is 47.4 ms and 156.7 ms. The receiver's sampling rate is also set to 250 KHz. The payload of each packet is 12 Bytes. To generate high network concurrency, we set the packet interval to 60 ms and 15ms for SF8 and SF10 packets respectively. Note that this setting is designed for generating the worst case scenario to test the scalability of the network. In other words, we use the setting to examine the maximum number of concurrent nodes that a network can support. Moreover, we use reverse engineering [18] to manipulate the payload of a packet, so that the first symbol denotes the packet ID. With the above settings, packets from different transmitters can be traced easily.

40 LoRa nodes are deployed on campus, as shown in Fig. 11(b). Referring to the figure, some LoRa nodes are placed inside two buildings. Others are placed along the main roads with pedestrians, cars and trucks passing by. When evaluating the performance of a n -node network, we randomly turn on n nodes. The gateway listens to the channel, and receives data when it is turned on. A round of experiment lasts 7 and 22 seconds with SF8 and SF10 respectively. In this case, each node transmits about 100 packets. For a given SF, results from three individual rounds of evaluation are collected.

B. Performance of Hi²LoRa

We studies and compares the performance of Hi²LoRa to Choir [14], CIC [6], PCube [8] and Magnifier [12]. The performance of two versions of Hi²LoRa are evaluated and presented here. The first version leverages the collision suppression scheme in the conference version of Hi²LoRa [21]; and the second version implement the enhanced collision suppression scheme mentioned in the previous section.

1) *Packet Detection and Synchronization Performance:* We first analyze the Packet Detection and Synchronization Rate (PDSR) for various schemes with SF8 and SF10 packets, respectively. Here PDSR refers to the percentage of packets being detected and synchronized correctly. Note that, a packet can be received correctly only if it is detected, and the first symbol of the payload is successfully identified. In addition, PDSR is evaluated with coarse-grained synchronization as coarse-grained synchronization is enough for detecting a packet.

Fig. 12 shows the PDSR for different schemes using SF8 and SF10 packets. The figure illustrates that both Hi²LoRa and PCube exhibit the highest packet detection rates compared to other schemes. Additionally, Hi²LoRa demonstrates a slightly higher detection rate than PCube under increased network concurrency. This is because PCube relies on SFD for synchronization, which may experience stronger interference from concurrent nodes. Nevertheless, both Hi²LoRa and PCube detect and synchronize up to 80% of packets when there are 40 concurrent nodes.

The PDSR of Magnifier is slightly lower than that of Hi²LoRa and PCube, e.g., up to 6.5% lower as the gateway detects

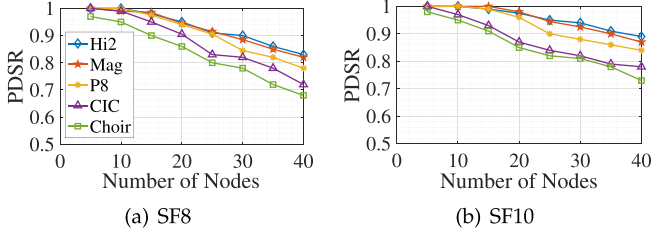


Fig. 12. Packet Detection and Synchronization Rate with different SF. Here "Hi²", "Mag", "PCube", "CIC" and "Choir" stand for the performance of Hi²LoRa with two antennas, Magnifier, PCube, CIC and Choir, respectively.

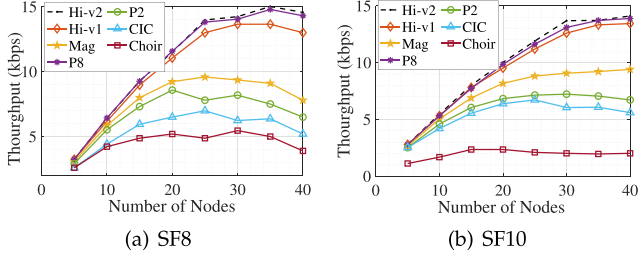


Fig. 13. Effective Throughput with different SF. In the figure, "Hi-v1" and "Hi-v2" refer to the first and the second version of Hi²LoRa respectively. "P2" and "P8" correspond to the performance of PCube with two and eight antennas, respectively.

and synchronizes to a transmitter with information from one single transmitting antenna. For CIC and Choir, the PDSR is significantly lower than that of the three schemes. For instance, the PDSR of CIC is up to 15.2% and 14.1% lower; and the PDSR of Choir is up to 22.1% and 21.9% lower than that of Hi²LoRa for networks with SF8 and SF10 packets, respectively. The primary reason stems from the fact that both schemes simply detect the arrival of a new packet by examining the presence of the preamble, without efficiently leveraging the features associated with the preamble symbols to combat interference, as a result, these schemes suffers more from co-channel interference caused by other concurrent packets.

2) *Effective Throughput Performance*: We next study the effective throughput of various schemes. Specifically, the effective throughput is calculated as the number of correctly received packet per second times the PHY layer payload size of these packets. The number of transmitting nodes in the system increases from 5 to 40. Recall that in networks with 40 nodes configured to transmit SF8 packets, there are on average 18 concurrent packets as duty cycle of each node is 0.44 (47.4 ms/(47.4 + 60) ms). For 40 nodes with SF10 packets, the average number of concurrent transmitters is 37 with duty cycle 0.91 (156.7 ms/(156.7 + 15) ms). Here, the duty cycle is set to a higher value than a normal LoRa node (e.g., usually configured with duty cycle equal to less than 1%). The high duty cycle is used to exam the maximum concurrency a network can support.

Fig. 13 shows the effective throughput of the network with different concurrency. The figure implies that both versions of Hi²LoRa achieve comparable performance to PCube with eight antennas. The enhanced version of Hi²LoRa (denoted by "Hi-v2") shows slightly higher throughput performance as it extracts

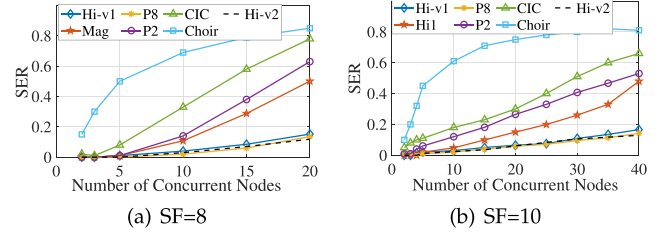


Fig. 14. Symbol Error Rate (SER) versus the number of concurrent packets.

more features to validate the correctness of the received symbols, thus helps to remove more interfering frequency tones from the received signal. Particularly, "Hi-v2" shows up to 12% and 8.7% higher effective throughput compared to that of "Hi-v1".

They also demonstrate significantly higher effective throughput than other schemes. For instance, with SF10 packets, the effective throughput of "Hi-v1" and "Hi-v2" is up to 42.9% and 50.0%, 140.0% and 152.0%, 566.7% and 600% higher than of Magnifier, CIC and Choir, respectively.

We also observe that the effective throughput does not scale with the number of concurrent nodes for networks with SF8 packets, but drops as network concurrency increases. As to networks with SF10 packet, if network concurrency keeps growing, similar performance degradation is expected. This is due to the fact that a packet is assumed to be received correctly only when all the data symbol can be correctly received. As a result, a packet will be dropped even though most of the symbols in it are correctly received.

3) *Concurrency Comparison*: This part studies the Symbol Error Rate (SER) performance with different numbers of concurrent nodes. Fig. 14 shows the average SER of the network with SF8 and SF10 packets. As expected, a lower SF (e.g., SF8) supports fewer concurrent packets than a larger SF (e.g., SF10). The reason is that the lower the SF, the lower the frequency resolution, and the higher frequency tone collisions each packet suffers.

In particular, for networks with SF8 packets, to keep SER ≤ 0.2 , the maximum number of concurrent nodes reaches 20 in PCube with eight antennas and both versions of Hi²LoRa; 10 in PCube with two antennas, 5 in CIC and 2 in Choir. For networks with SF10 packets, the number of concurrent nodes approaches 40 in PCube with eight antennas and Hi²LoRa; 15 in PCube with two antennas, 10 in CIC and 3 in Choir. The excellent performance of Hi²LoRa reveals its ability to 1) identify different packets with sufficient features; and 2) suppress most frequency tone collisions with only two receiving antennas.

4) *Complexity Comparison*: Since the performance trends for SF8 and SF10 packets are similar, we only discuss network performance with SF10 packets in the rest of the paper. To analyze and compare the complexity of various schemes, we run them on a computer with 2, 4, and 8 concurrent packets. The decoding time is normalized to a standard LoRa decoder. Fig. 15 shows the normalized decoding complexity.

As implied in the figure, the enhanced version of Hi²LoRa achieves a 20% to 35% reduction in computation time compared to the original, by efficiently excluding interfering frequency tones (typically side-lobes from other packets) at an earlier stage.

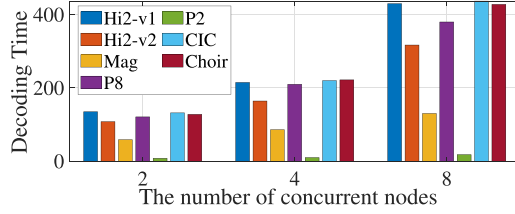


Fig. 15. Complexity Comparison for various scheme with SF = 10.

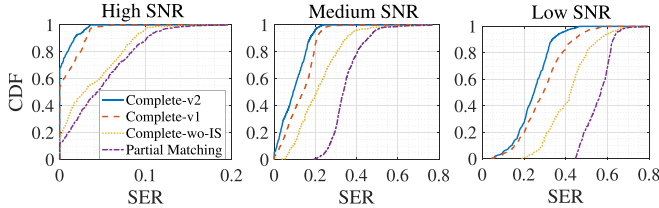


Fig. 16. Symbol Error Rate distributions after different modules. Here SF=10. The legend "Partial-Matching" represents the CDF of SER after the Partial-Matching module. "Complete-wo-IS", "Complete-v1", and "Complete-v2" refer to the CDF of SER after the Complete-Matching module without and with both versions of collision suppression algorithms, respectively.

This eliminates the need to extract fine-grained offset information from these interfering tones, saving additional processing time. Moreover, the complexity of both version of Hi²LoRa is higher than that of PCube with two antennas. However, it is comparable to PCube with eight antennas, CIC and Choir. We argue that the complexity of Hi²LoRa is not a problem as the decoder is implemented in a gateway with sufficient computing resources and power supplies. Moreover, it usually has easy access to a powerful edge cloud that can help run the decoder.

Note that there is no additional energy consumption introduced for each LoRa transmitter. The energy consumption associated with sending a packet from a transmitter, whether in Hi²LoRa or most existing methods, remains equivalent to that of a standard LoRa radio. Furthermore, since gateways generally have access to ample power supply, rendering energy consumption a non-limiting factor. We thus believe that energy consumption is not a critical concern in our proposed scheme.

5) *Performance Improvement of Different Modules*: The performance of packets passing through different modules is examined here. We investigate performance improvements for packets with different SNRs here. The packets are first categorized into low (with $-10 \text{ dB} \leq \text{SNR} \leq 0 \text{ dB}$), medium (with $0 \text{ dB} \leq \text{SNR} \leq 10 \text{ dB}$) and high (with $\text{SNR} \geq 10 \text{ dB}$) SNR packets. According to the topology, there are 25%, 50%, and 25% of low, medium, and high SNR packets. Fig. 16 plots the CDF of a packet's SER after signals processed via Partial-Matching, Complete-Matching without collision suppression (denoted by "Complete-wo-IS"), and Complete-Matching with two versions of collision suppression schemes (denoted by "Complete-v1" and "Complete-v2"). According to the figure, after Complete-Matching with collision suppression, the SER is improved significantly.

Specifically, for low SNR packets, the thresholds for identifying symbols such as Γ_δ , Γ_ϕ , Γ_τ , and Γ_a (listed in Table II)

are larger than those of medium and high SNR packets, leading to higher decoding ambiguity, and higher SER. In this case, efficient complete matching and collision suppression algorithms are crucial to resolve the decoding ambiguity. Refer to Fig. 16, with Partial-Matching, the SER of 80% of the packets is lower than 0.61, and this value is reduced to 0.48, 0.38 and 0.31 when performing Complete-Matching without and with collision suppression v1 and v2, respectively. Compared to the partial-matching module, on average the SER improvements of Complete-Matching without and with collision suppression v1 and v2 is 32.0% 60.1% and 70.6%, respectively. Overall, low SNR packets benefit the most from Hi²LoRa, as they are more susceptible to both noise and interference.

For medium SNR packets, the SER of 80% of the packets is lower than 0.40, 0.31, 0.20 and 0.155 for Partial-Matching, Complete-Matching without collision suppression, and with suppression algorithm v1 and v2, respectively. Compared to Partial-Matching module, the average SER improvement after Complete-Matching without collision suppression and with collision suppression v1-and v2 is 20.4% and 33.8%, and 39.0%, respectively.

As to high SNR packets, the small thresholds reduce decoding ambiguity. Therefore, collision suppression plays a more pivotal role in decoding a packet. In particular, the SER of 80% of the packets is lower than 0.091, 0.082, 0.017 and 0.009 for Partial-Matching, Complete-Matching without collision suppression, and with collision suppression v1-and v2, respectively. Moreover, all high SNR packets can be received with SER lower than 4.5% when processed by the Complete-Matching module with collision suppression.

VII. RELATED WORKS

A. Distributed MIMO

Multiple receiving antennas receive concurrently packets from different senders and form a distributed MIMO system. Distributed MIMO has been extensively studied in LTE, Wi-Fi, mesh networks and IoT systems [15], [16], [22], [23]. Since they require complicated channel estimation and joint decoding for all the transmitters, such approaches incur prohibitively high implementation complexity when applying to LoRaWAN. Moreover, all the schemes do not integrate the characteristics of CSS, hence are inefficient in LoRa concurrent decoding. PCube [8] inspects the channel differences between antennas for parallel decoding. To get satisfactory performances, up to eight antennas are needed, rendering an expensive solution.

B. Concurrent Reception in LoRaWAN

1) *MAC Layer Approaches for Collision Avoidance*: Extensive work has studied how to avoid packet collisions in LoRaWAN [24]. Prior work [25], [26], [27], [28] set different SF, bandwidth, and time slots for different users to eliminate transmission collisions. LMAC [29] and prior work [30] propose to detect preamble chirp to achieve reliable carrier sense based on the Channel Activity Detection (CAD) mechanism in LoRa radios. Still, implementing CAD requires additional firmware

changes. Besides, the MAC layer approaches cannot wholly resolve collisions as long as the transmitters are far from each other and thus cannot sense the transmissions from others.

2) *PHY Layer Solutions for Concurrent LoRa Reception:* Choir [14] ignores the existence of STO and SFO, and assumes a steady CFO for the entire packet. However, the location of the frequency peak (assumed to be the CFO of a packet) changes with time due to SFO. Assuming a fixed frequency drift makes the receiver quickly lose track of the actual location of the peak, resulting in packet reception failure. Regarding CIC [6], it employs the frequency drift and the time misalignment between different packets. Similar to Choir, as the extracted CFO is inaccurate, CIC's performance degrades significantly when network concurrency increases. Magnifier [12] proposes to use the CFO, STO and SFO for parallel decoding in LoRaWAN. However, as Magnifier only uses one receiving antenna, the dimension of its features is limited. Moreover, the single antenna system in Magnifier is not able to suppress frequency tone collisions. As a consequence, the number of concurrent transmitters Magnifier can support is significantly lower than that of Hi²LoRa.

By ignoring all the offsets, FTrack [5], CoLoRa [7], SCLoRa [9], NScale [10], OTC [31], Pyramid [11], and AlignTrack [32] exploit the fact that different packets arrive at the receiver at different time. FTrack [5] leverages the frequency continuity for parallel decoding. OTC [31] introduces an online algorithm to process concurrent packet with the packet arrival time misalignment information. Besides the different arrival times, Pyramid [11], AlignTrack [32], and SCLoRa [9] further adopt the average energy of each packet for parallel decoding. Linear scaling is applied to scale different chips in CoLoRa [7]. The energy distributions of the scaled frequency bins in two successive symbols are examined to map frequency bins to packets. With limited number of inaccurate features, and the lack of an efficient strategy for suppressing frequency tone collisions, the above schemes can only be applied to low concurrency networks.

Successive Interference Cancellation (SIC) is discussed in mLoRa [33] for jointly decoding multiple packets. Unfortunately, neglecting the offset impacts on the received signal leads to substantial decoding errors. According to [33], mLoRa can support two to three concurrent transmissions, which shows significantly lower efficiency than Hi²LoRa.

NScale [10] and CurvingLoRa [34] leverage non-linear scalings for separating concurrent packets. Extensive hardware modifications are needed as non-linear scaling changes the modulation of all the LoRa nodes. Netscatter [35] and P²LoRa [36] enables large-scale concurrent transmissions with backscatter devices. The schemes only work for short-range communications due to the devices' extremely low energy. It also requires strict synchronization of all transmitters, which is unsuitable for distributed implementation.

VIII. CONCLUSION

This work proposes Hi²LoRa, a framework that utilizes highly dimensional and highly accurate features for LoRa concurrent decoding. Hi²LoRa leverages both hardware imperfections and

channel state information inherent to each transceiver for packet identification. The accuracy of the high dimensional features is achieved by a two-stage matching algorithm discussed in Hi²LoRa, as well as its ability to suppress frequency tone collisions. The extensive results demonstrate that Hi²LoRa achieves a good implementation complexity and performance trade-off. For instance, it either shows comparable concurrency to that of PCube with eight antennas or achieves $2.7\times$ of concurrency compared to PCube with the same number of antennas (e.g., two antennas). Additionally, Hi²LoRa provides up to $3.5\times$ higher concurrency than most state-of-the-art work such as CIC [6] with one antenna. Hi²LoRa can also be extended to resolve non-orthogonal concurrent packet receptions. For instance, receive concurrently transmitted packets with SF6 bandwidth=125 MHz; SF8 bandwidth = 250 MHz; and SF10, bandwidth = 500 MHz. In future work, the authors plan to explore efficient collision suppression schemes with massive receiving antennas to further extend network concurrency.

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Weiwei Chen received the PhD degree from the Hong Kong University of Science and Technology, Hong Kong, in 2013. She is now with the Department of Computer Engineering and Science, Shanghai University. Prior to that, she was with the Department of Computer Science and Engineering, Southeast University, China. Her research interests include wireless sensing and localization, IoT networking, cross technology communications (CTC), and mobile edge computing.



Xianjin Xia (Member, IEEE) received the BS, MSc and PhD degrees in computer science from Northwestern Polytechnical University, Xi'an, China, in 2010, 2013, and 2018, respectively. He is currently a research assistant professor (RAP) with the Department of Computing, the Hong Kong Polytechnic University. His research interests include the Internet of Things, low-power wide-area network, wireless localization and tracking, and mobile computing. He is a member of ACM.



Shuai Wang received the BS and MS degrees from the Huazhong University of Science and Technology, China, and the PhD degree from the Department of Computer Science and Engineering, University of Minnesota, in 2017. He is currently a professor and department head with the Department of Computer Engineering, Southeast University. His research interests include the wireless networks and sensors, cyber-physical systems, and data science.



Tian He (Fellow, IEEE) is currently a professor with the Department of Computer Science and Engineering, University of Minnesota Twin Cities. He is the author or coauthor of more than 280 papers in premier network journals and conferences with more than 22000 citations (H-Index 65). His research interests include wireless networks, networked sensing systems, cyber-physical systems, real-time embedded systems, and distributed systems. He was the recipient of the NSF CAREER Award in 2009, the McKnight Land-Grant Chaired Professorship in 2011, the China NSF Outstanding Overseas Young Researcher I and II, in 2012 and 2016, the George W. Taylor Distinguished Research Award, in 2015, and seven best paper awards in international conferences, including MobiCom, SenSys, and ICDCS. He has served a few general/program chair positions in international conferences and on many program committees and also has been an editorial board member for six international journals. He is an ACM fellow.



Shuai Wang is currently an assistant professor with the School of Computer Science and Engineering at Southeast University in China. He is broadly interested in wireless sensing, mobile computing, wireless communication, IoT, and embedded systems for improving wireless connectivity. He won the Best Paper Award at ICDCS 2018.



Gang Liu received the PhD degree in computer science from Hunan University, China, in 2021. He conducted his postdoctoral research with Hunan University from 2021 to 2023 and now holds an associate professor position with Shenzhen Institute for Advanced Study, UESTC. His research focuses mainly on security, high performance computing, cloud computing, and computer system architecture.



Caishi Huang received the MPhil and PhD degrees in electronic and computer engineering from the Hong Kong University of Science and Technology, in 2007 and 2012 respectively. He is the associate master with Cheong Kun Lun College and assistant professor with the Department of Biomedical Science, Faculty of Health Science, University of Macau. His general research interests include distributed networks, cellular and computer networks, multimedia, artificial intelligence, and Big Data.